

Sharing the Keys: A Tangible Research Prototype for Exploring Shared Financial Decision-Making in Caregiving

Bárbara Maria Martins da Silva

Department of Design
The Ohio State University
Columbus, Ohio, USA
dasilva.48@osu.edu

Abstract

As populations age, financial decision-making is increasingly shaped by relationships between older adults and caregivers, rather than by individual account holders acting alone. Yet most digital financial services are designed around an individual mental model of financial control, assuming a single user per account. Sharing the Keys is a tangible research prototype designed as an interaction probe to explore how people reason about shared financial decision-making in caregiving contexts. The prototype combines physical artifacts and role-playing to externalize financial tasks and prompt collaborative reflection. Insights from exploratory workshops show how tangible interaction and role-playing supported perspective-taking and surfaced relational concerns around assistance, autonomy, and financial exploitation. Rather than proposing a finalized solution, this work demonstrates the value of early, pre-product co-design for understanding the dynamics that shape real financial practices and for informing future account structures and user experiences in digital financial services for aging populations.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Interaction design; Interaction design process and methods; Participatory design.

Keywords

Financial caregiving, Shared financial decision-making, Aging, Research through design

ACM Reference Format:

Bárbara Maria Martins da Silva. 2026. Sharing the Keys: A Tangible Research Prototype for Exploring Shared Financial Decision-Making in Caregiving. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799302>

1 Introduction

As people age, managing finances often becomes a shared, relationship-based task between older adults and their caregivers, including family members, friends, or professionals. Financial management is considered an Instrumental Activity of Daily Living (IADL) and a key indicator of functional independence [4]. Beyond

practical support, shared financial decision-making involves ongoing emotional and ethical negotiations around trust, autonomy, and responsibility. Caregiving, in this sense, is a relational process shaped by continuous negotiation of control and identity rather than a simple transfer of tasks [5].

Despite this interdependence, most digital financial systems continue to assume a single, autonomous user, reflecting an individual mental model of financial control. When financial responsibilities must be shared, older adults and their supporters often rely on informal and insecure workarounds, such as sharing passwords or credentials, to maintain access and oversight [1]. Recent HCI research has documented these practices as forms of “unofficial proxy” support, in which caregiving assistance emerges outside formal system boundaries due to limited support for shared access and negotiated responsibility [2]. While work on proxy accounts and delegated access has begun to address these issues [3], many financial technologies still struggle to support partial authority, ongoing communication, and collaborative decision-making.

Addressing these challenges requires designers to adopt approaches that allow people to reflect on shared financial practices before committing to specific technological solutions. Participatory and tangible design methods have been shown to support reflection and collaboration around sensitive topics by externalizing abstract or emotionally charged issues [6].

In this paper, we present Sharing the Keys, a tangible research prototype designed as an interaction probe to explore shared financial decision-making in caregiving contexts. Rather than proposing a solution, the prototype combines physical artifacts and role-playing to support early, pre-product exploration of how people reason about assistance, autonomy, and risk, surfacing interaction challenges that are often obscured in digital financial interfaces.

2 RESEARCH GOAL AND APPROACH

The goal of this work is to explore how people reason about shared financial decision-making in caregiving contexts, with particular attention to how autonomy, assistance, and risk are negotiated. Rather than evaluating a system or validating requirements, we adopt a Research Through Design (RtD) approach in which a tangible prototype is used as a means of inquiry.

Drawing on participatory design traditions, we position Sharing the Keys as an interaction probe that encourages reflection, discussion, and sense-making around sensitive financial topics. Following Sanders and Stappers [5], the prototype is not intended to produce generalizable findings, but to surface experiential and relational insights that can inform future financial system and interaction design. By using a physical artifact to explore challenges associated



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ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3799302>

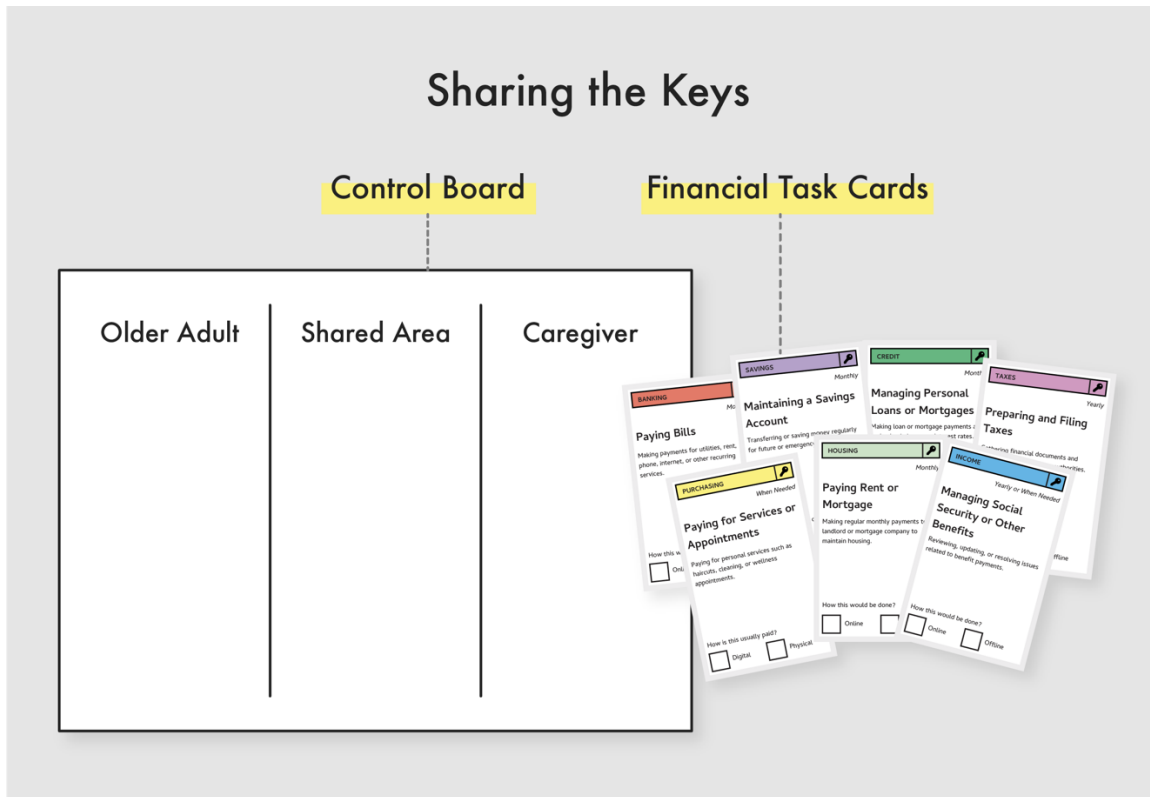


Figure 1: The Sharing the Keys prototype consists of a physical board and task cards for negotiating levels of financial responsibility.

with digital financial services, we aim to make abstract issues of access, control, and responsibility more tangible and discussable in early-stage design exploration.

3 THE SHARING THE KEYS PROTOTYPE

Sharing the Keys is a tangible research prototype designed to mediate conversations about how financial tasks are shared between older adults and caregivers (Figure 1). The prototype consists of a physical board divided into control zones (older adult, shared area, and caregiver), along with task cards (the “keys”) representing common financial activities such as paying bills, managing accounts, tracking expenses, or monitoring transactions. The cards are color-coded to represent different task categories, including banking, purchasing, savings, and taxes.

Participants interact with the prototype by placing and moving task cards across the control zones. The act of positioning a task serves as a prompt for discussion, negotiation, and reflection, allowing participants to articulate concerns, assumptions, and expectations related to financial decision-making. As part of the interaction, participants also indicate whether tasks would be carried out online or offline, encouraging reflection on how financial responsibilities intersect with digital systems. The physical nature of the prototype supports the externalization of decisions that are

often emotionally charged or difficult to verbalize, shifting attention from interpersonal confrontation to shared interaction with artifacts.

The prototype was intentionally designed to remain open-ended. It does not prescribe correct configurations or optimal outcomes, but instead supports exploration of multiple possibilities and tensions. In this way, Sharing the Keys functions as a research instrument rather than a solution, enabling early inquiry into shared financial practices without committing to specific interface designs or interaction models.

4 EXPLORATORY WORKSHOPS

The prototype was explored through two workshop sessions conducted in an academic setting. Participants were undergraduate students who engaged in role-playing to step into the perspectives of older adults and caregivers. While participants did not have lived experience in these roles, role-playing was used to encourage perspective-taking and to surface relational dynamics that extend beyond individual preferences.

At the beginning of each session, participants were paired and assigned scenario cards describing a caregiving situation involving a common challenge in older adult–caregiver dynamics. These scenarios outlined contextual constraints and tensions, such as an older adult who is motivated to learn digital tools but requires assistance, paired with a caregiver who has limited time or availability.



Figure 2: Participants interacting with the Sharing the Keys prototype during role-playing activities.

Scenario cards were used to situate participants in a shared context and to guide role adoption.

After roles were assigned, pairs interacted with the Sharing the Keys prototype to discuss how financial tasks could be divided between the two roles (Figure 2). Participants negotiated task placement across control zones in an effort to balance autonomy, assistance, and perceived risk. Following this activity, pairs identified challenges and tensions that emerged from their task distributions, such as uncertainty around trust, security concerns, responsibility boundaries, or communication breakdowns.

Each session concluded with a group discussion and a speculative design activity. Participants shared their scenarios and reflections with the larger group, then created poster concepts imagining how digital financial services might better address the challenges surfaced during role-playing. These posters were used as reflective and generative artifacts rather than as proposed solutions. Data sources for analysis included observations of role-playing interactions, written reflections collected after the sessions, and the content of the speculative poster concepts.

The workshops were exploratory in nature and not intended to evaluate the prototype or assess proposed solutions. Instead, they were designed to surface interaction patterns, concerns, and values related to shared financial decision-making, informing early understanding of how caregiving relationships shape financial practices.

5 INSIGHTS FROM USING THE PROTOTYPE AS AN INTERACTION PROBE

The insights below reflect two complementary dimensions: how the prototype shaped participant behavior during exploration, and what discussions revealed about caregiving and financial decision-making when viewed through role-playing and speculation.

5.1 What the toolkit enabled when exploring sensitive financial topics.

5.1.1 Externalizing negotiation through tangible interaction. During role-playing, participants frequently used the physical task cards and board zones as mediators in conversation. Rather than directly asserting opinions, participants pointed to cards, moved them hesitantly, or paused before placing them, often verbalizing uncertainty or seeking agreement. This behavior was especially evident when discussing tasks associated with higher risk, such as banking access or payments. Participants described the activity as making financial decisions “easier to talk about,” suggesting that tangible interaction helped externalize sensitive negotiations into shared sense-making.

5.1.2 Perspective-taking through role-playing. Role-playing encouraged participants to reason from caregiving and older adult perspectives rather than their own. Participants referenced fears of losing dignity, being blamed, or not understanding digital systems when explaining task placement. When scenarios included clearer

constraints such as limited caregiver time or fear of online banking, discussions shifted from abstract opinions to negotiated understanding of relational boundaries.

5.2 What discussions revealed about caregiving and financial decision-making.

5.2.1 Assistance as mediated rather than delegated. Speculative poster concepts consistently framed assistance as a mediated process rather than a full transfer of control. Participants proposed designs involving shared visibility, differentiated permissions, and confirmation steps, where caregivers assist without fully taking over financial authority. Advisory roles, human or AI-based, were also imagined as intermediaries supporting decision-making rather than replacing it.

5.2.2 Security and fear of irreversible consequences. Concerns about scams, theft, and irreversible mistakes were prominent across discussions and speculative designs. Participants emphasized safeguards such as confirmations, alerts, and monitoring, reflecting anxieties about loss of control and financial exploitation. These concerns suggest that shared financial decision-making is shaped not only by usability, but by emotional and relational dimensions of security.

6 DISCUSSION AND IMPLICATIONS FOR HCI

The findings from Sharing the Keys point to several implications for HCI research and the design of financial technologies. First, they highlight the limitations of single-user assumptions in digital financial services and the need for interaction models that support negotiated, shared responsibility. These findings suggest that future systems could benefit from role-dependent permissions, accountability, and communication around shared financial tasks.

Second, the prominence of safeguards and mediated assistance underscores the tension between autonomy and protection in caregiving contexts. Prior work on unofficial proxy practices in banking shows how assistance often emerges through informal negotiation when systems fail to support shared access [2]. This work complements this literature by using an early-stage interaction probe to surface how such negotiations are reasoned about before access arrangements are formalized.

Finally, this work demonstrates the value of pre-product co-design as an interaction inquiry. By engaging participants before a final system is defined, the prototype enabled reflection on underlying relational challenges rather than feature-level evaluation. For sensitive domains such as financial caregiving, tangible and role-based prototypes can play a critical role in revealing concerns that may otherwise remain implicit.

7 LIMITATIONS AND FUTURE WORK

This work is limited by its exploratory scope and the use of undergraduate participants engaging in role-playing rather than lived caregiving experiences. While this approach enabled diverse perspectives to emerge, future work will apply the prototype in real caregiving contexts with older adults and caregivers. Further iterations of the prototype may also explore long-term dynamics of consent, accountability, and changing roles over time.

8 CONCLUSION

Sharing the Keys demonstrates how a tangible research prototype can function as an interaction probe to explore shared financial decision-making in caregiving contexts. By externalizing sensitive tasks and supporting perspective-taking, the prototype surfaced interaction challenges related to autonomy, assistance, and risk that are often obscured in digital financial interfaces. We present this work as an invitation for feedback and discussion within the CHI community, with the goal of informing future research and design of more inclusive financial technologies for aging populations.

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